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ADVANCE MESSAGING SYSTEM

HLR LOOKUP

HLR LOOKUP API DOCUMENT

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HLR LOOKUP API

➢ Introduction:

The **Home Location Register (HLR)** is the main database of permanent subscriber information for a mobile network. The **HLR** is an integral component of CDMA (code division multiple access), TDMA (time division multiple access), and GSM (Global System for Mobile communications) networks.

➢ Request Methods: System Requests can be made by following two methods:

- SMPP
- HTTP

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SMPP

The Short Message Peer-to-Peer (**SMPP**) is a protocol used by the telecommunications industry for exchanging SMS messages between Short Message Service Centers (SMSC) and External Short Messaging Entities (ESME). The protocol is a level-7 TCP/IP protocol, which allows fast delivery of SMS messages.

The connection between application and AMS SMPP Server is SMPP version 3.4

Using SMPP Account, It is possible to Request for Number Lookup. In order to Use SMPP Lookup Service, User must have valid systemid & password to Set in Bind PDU.

BIND Operation

The purpose of the SMPP bind operation is to register an instance of an ESME with the SMSC system and request an SMPP session over this network connection for the submission or delivery of messages. Thus, the Bind operation may be viewed as a form of SMSC login request to authenticate the ESME entity wishing to establish a connection.

Requested Parameters:

System_id = Provided System Id

Password = Provided Password

System_type = **HLR or MNP**

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BIND Mode

User may bind as either Tx (Transmitter), Rx (Receiver) or TRx (Transceiver).

Tx: User bound as a Transmitter is authorized to send short messages and to receive the corresponding SMPP responses.

Rx: User bound as a Receiver is authorized to receive short messages and to return the corresponding SMPP message responses.

TRx: User bound as a Transceiver is allowed to send & receive messages over a single SMPP session.

Users are allowed to bound as only 1 TRx/Rx with multiple Tx Session.

SUBMIT Operation

SubmitSM PDU is used for submitting the number context Request, having destAddress parameter set to the required destination address. All other parameters (srcAddress, esm, ton/npi) will be ignored.

submitSMResp PDU will be used as Response containing responseid (Unique Number to Identify Request) on Successful Submission or Submit_Error in case of any Error. The *command_status* field of an SMPP message response indicates the success or failure of an SMPP request.

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SMPP Error Codes and their associated values are defined in the following table:

Code	Name	Description
0x00000000	ESME_ROK	No Error
0x00000003	ESME_RINVCMDID	Invalid Command ID
0x00000008	ESME_RSYSERR	System Error
0x00000010	ESME_RINSBAL	Insufficient Balance
0x0000000B	ESME_RINVDSTADR	Invalid Destination Address
0x0000000D	ESME_RBINDFAIL	Bind Failed
0x0000000E	ESME_RINVPASWD	Invalid Password
0x0000000F	ESME_RINVSYSID	Invalid System ID
0x00000012	ESME_RCOVERR	No Coverage For Requested Destination
0x00000016	ESME_RACCEXPIRY	Account Expired

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Lookup Result

Once the Lookup request is being finalized on our System, **deliverSM** PDU is sent as a lookup Result containing the Status & other required information included in the *short_message* field of the deliverSM for the requested destination number.

Example (System_type=HLR):

(deliver: (pdu: 224 5 0 5001) (addr: 0 0 HLR) (addr: 0 0 919589xxxxxx) (sm: msg: id:6002 sub:001 dlvrd:001 submit date:1507140141 done date:1507140142 stat:DELIVRD err:000 imsi: 404939383926417 msc: 919893 cc:91 nnc:40493 isPorted:false p_nnc: isRoaming:false r_cc: r_nnc: text:dlr) (opt: (str: (tlv: 30) 6002)))

If Number is INVALID/ABSENT_SUBSCRIBER etc, The DeliverSm will be as follow:

(deliver: (pdu: 224 5 0 5000) (addr: 0 0 HLR) (addr: 0 0 919589xxxxxx) (sm: msg: id:6000 sub:001 dlvrd:001 submit date:1507141154 done date:1507141225 stat:UNDELIV err:011 imsi: msc: cc:91 nnc:40493 isPorted:false p_nnc: isRoaming:false r_cc: r_nnc: text:dlr) (opt: (str: (tlv: 30) 6000)))

Example (System_type=MNP):

(deliver: (pdu: 224 5 0 5001) (addr: 0 0 HLR) (addr: 0 0 919589xxxxxx) (sm: msg: id:6002 sub:001 dlvrd:001 submit date:1507140141 done date:1507140142 err:000 cc:91 nnc:40493 isPorted:false p_nnc: isRoaming:false r_cc: r_nnc: text:dlr) (opt: (str: (tlv: 30) 6002)))

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If Number is INVALID/ABSENT_SUBSCRIBER etc, The DeliverSm will be as follow:

(deliver: (pdu: 224 5 0 5000) (addr: 0 0 HLR) (addr: 0 0 919589xxxxxx) (sm: msg: id:6000 sub:001 dlvrd:001 submit date:1507141154 done date:1507141225 err:011 cc:91 nnc:40493 isPorted:false p_nnc: isRoaming:false r_cc: r_nnc: text:dlr) (opt: (str: (tlv: 30) 6000)))

Here **err: 011** is the status code of the Lookup. See the status code table for description.

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HTTP

Users can Use Following Services via HTTP API:

1) Number Lookup
2) Status Enquiry

1) Number Lookup

Number Lookup HTTP API supports both GET and POST Methods. Using HTTP Method limit of upto 100 numbers is allowed in a single request.

Description:

1.1 URL For Lookup Request:
http://ip:port/websmpp/hlr/lookup?username=xxxx&password=xxxx&destination=919589xxxxxx

1.2 URL For Lookup Request and Result immediately:
http://ip:port/websmpp/hlr/lookupandresult?username=xxxx&password=xxxx&destination=919589xxxxxx

Requested Parameters:

username = Provided System Id

Password = Provided Password

Destination = Destination Number **or** Comma separated Destination Numbers

Response Format

On Successful Lookup:

BatchId: 140715134158696

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On Failed:

ERROR - HTTP01 --> Invalid HTTP URL Format

2) Status Enquiry:

HLR: After Successful Lookup, User can get Result via Following URL:

http://ip:port/websmpp/hlr/report?batchid=140715134158696

Result Format: JSON (JavaScript Object Notation):

{"cc":"91","nnc":"404093","isRoaming":"No","p_nnc":"404078","isPorted":"Yes","destination":"91958905XXXX","error_code":"000","id":"23112715404110001","time":"2023-11-27 15:40:41","batchid":"27112315404063977 67","error":"NO_ERROR", "msc":"3333","status":"DELIVRD","imsi":"9374747474" }

MNP: After Successful Lookup, User can get Result via Following URL:

http://ip:port/websmpp/mnp/report?batchid=140715134158696

Result Format: JSON (JavaScript Object Notation):

{"cc":"91","nnc":"404093","isRoaming":"No","p_nnc":"404078","isPorted":"Yes","destination":"91958905XXXX","error_code":"000","id":"23112715404110001","time":"2023-11-27 15:40:41","batchid":"27112315404063977 67","error":"NO_ERROR"}

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Parameter Description:

Id : Unique Id to Identify this request
batchid: Batch id provided at request time
destination: Requested Destination Number
time: lookup request time
error_code: error code if lookup failed/number error
error: error description if lookup failed/number error
cc: Country code of destination number
nnc: Network code
isPorted: Yes/No
p_nnc: ported network code if ported
isRoaming: Yes/No
r_cc: Roaming Country code if roaming
r_nnc: Roaming Network code if roaming

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Status Code Table (1)

This section Describes Errors Returned as Response On Submission (if Failed):

Code	Name	Description
001	INVALID_LOGIN	Either Systemid or password is incorrect
002	INVALID_REQUEST	Required parameters are missing or Invalid parameters or invalid values of parameters
003	ACCOUNT_EXPIRED	The Account has been Expired
004	INSUF_BALANCE	Insufficient Balance to fulfill the Request
005	INVALID_DEST_ADDR	Invalid Destination Address
006	NO_COVERAGE	No Coverage for Requested Destination
007	SYSTEM_ERROR	System Error Occurred

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Status Code Table (2)

This section describes Status of the Lookup Result returned by the system.

Code	Name	Description
011	ABSENT_SUBSCRIBER	The Subscriber is absent
012	UNKNOWN_SUBSCRIBER	The subscriber is unknown
013	UNIDENTIFIED_SUBSCRIBER	The Subscriber is unindentified
014	TIMEOUT	Timeout occurred
015	PROVIDER_GENERAL_ERROR	General Error From Provider
016	NO_RESPONSE	No Status From Provider
017	SYSTEM_FAILURE	A system failure occurred.
018	INVALID_MSISDN	Invalid msisdn
019	CALL_BARRED	Call Barred
020	ILLEGAL_SUBSCRIBER	Illegal Subscriber
021	TELESERVICE_NOT_PROVISIONED	Teleservice not provisioned
022	FACILITY_NOT_SUPPORTED	No facility
023	SUBSCRIBER_BUSY_FOR_MT_SMS	Subscriber Busy
024	UNEXPECTED_DATA_VALUE	Unexpected data
025	ABSENT_SUBSCRIBER_TMP	Temporary Absent
026	DELIVERY_FAILURE	Delivery failed

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